DOCUMENTATION FOR C++ STRUCTURE AND POINTER EXAMPLE

This program demonstrates how to use **structures** and **pointers** in C++. It defines a `Time` structure to represented in hours, minutes, and seconds and shows how to manipulate its members using both direct access and pointer dereferencing.

1. Structure Definition

```
struct Time {
    int hour;    // 0-23
    int minute;  // 0-59
    int second;  // 0-59
};
```

- The `Time` structure contains three integer members: `hour`, `minute`, and `second`.
- It models a simple clock time.

2. Function Definition

```
void printTime(Time t) {
    cout << t.hour << " : " << t.minute << " : " << t.second << endl;
}
```

- `Print Time` accepts a `Time` object and prints its values in the format `hour: minute: second`.
- This function improves readability by separating printing logic from `main ()`.

3. Main Function

```
int main() {
    Time lunchTime;                      // Declaration of Variable lunchTime
    Time dinnerTime = {18, 30, 0}; // Declaration & Initialization of Variable dinnerT
    Time *ptrTime = &lunchTime;     // Declaration & Initialization of Pointer Variable

    lunchTime.hour = 12;            // Set member variable hour of lunchTime
    ptrTime->minute = 30;           // Set member variable minute of lunchTime
    (*ptrTime).second = 20;         // Set member variable second of lunchTime

    cout << "Lunch will be held at ";
    printTime(lunchTime);

    cout << "Dinner will be held at ";
    printTime(dinnerTime);

    return 0;
}
```

- **Variable Declaration:**
  - ✓ `lunchTime` is declared but not initialized.
  - ✓ `dinnerTime` is declared and initialized to **`18:30:00`**.
- **Pointer Usage:**
  - ✓ `ptrTime` points to `lunchTime`.
  - ✓ Members of `lunchTime` are set using both direct access (`lunchTime.hour`) and pointer dereferencing (`ptrTime->minute`, `(*ptrTime).second`).